Statistics in Psychology and Education

Course Syllabus

Instructor: Hau-Hung Yang

Course Information

Course Level: Undergraduate
Credits: 3 credits (one semester)
Prerequisites: None
Software: R (with RStudio)

Course Description

This course provides psychology and education students with a comprehensive introduction to applied statistical methods commonly used in psychological and educational research. The course emphasizes intuition, correct interpretation, and hands-on data analysis with R. Students will learn to formulate research questions in statistical terms, choose appropriate analytic strategies, conduct analyses, and critically evaluate statistical results reported in published research. By the end of the course, students will possess the practical skills to analyze data independently and the conceptual understanding to be informed consumers of quantitative research.

Course Objectives

Upon completion of this course, students will be able to:

1. Understand the logic of statistical inference, including estimation and hypothesis testing.
2. Select appropriate statistical methods for common research designs in psychology and education.
3. Conduct statistical analyses using R and interpret output correctly.
4. Critically evaluate the use of statistics in published research articles.
5. Communicate statistical results clearly in written reports.

Textbook and References

- Howell, D. C. (2013). *Statistical Methods for Psychology* (8th ed.). Cengage Learning. **[Primary textbook]**
- Field, A., Miles, J., & Field, Z. (2012). *Discovering Statistics Using R*. SAGE Publications. [Supplementary reference]

Assessment

Homework assignments (weekly R exercises)	30%
Midterm examination	25%
Final examination	25%
In-class quizzes and participation	10%
Data analysis project	10%

Weekly Schedule

Week	Topic
1	Introduction to statistics: Why do we need statistics? Populations, samples, and the logic of inference; getting started with R and RStudio
2	Descriptive statistics and data visualization: Central tendency and variability; histograms, boxplots, and scatterplots in R
3	Probability foundations: Basic probability rules, conditional probability, Bayes' theorem
4	Random variables and probability distributions: Discrete and continuous distributions; the normal distribution
5	Sampling distributions and the Central Limit Theorem
6	Point estimation and interval estimation: Confidence intervals for means
7	Introduction to hypothesis testing: Null and alternative hypotheses, Type I and Type II errors, p-values
8	*Midterm Examination*
9	One-sample and two-sample t-tests
10	Paired-samples t-test; statistical power and effect size
11	One-way analysis of variance (ANOVA): Between-subjects designs
12	Multiple comparisons and post-hoc tests

Week	Topic (continued)
13	Correlation: Pearson and Spearman correlation coefficients
14	Simple linear regression: Model, estimation, and interpretation
15	Chi-square tests for independence and goodness of fit; review: choosing the right test and common misuses of statistics
16	*Final Examination*

The schedule follows the 16-week NTU semester: fourteen teaching weeks, the midterm examination in Week 8, and the final examination in Week 16. A class lost to a national holiday is made up in the flexible week after Week 16 (Week 16+1), as the NTU academic calendar allows.

Course Policies

- All homework must be submitted as reproducible R scripts or R Markdown documents.
- Late submissions will incur a 10% penalty per day, up to a maximum of three days.
- Academic integrity is expected; all individual work must be the student's own.
- Students are encouraged to use AI tools (e.g., ChatGPT) for learning, but must demonstrate understanding of the output and clearly cite any AI-assisted work.